Dewen Zeng

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RESEARCH INTEREST

My research is focusing on **data miming and embeded computing**.

Key words: Internet of Things, Embedded Systems, Deep Learing, Hardware Acceleration.

EDUCATION

09/2016 – Present	Huazhong University of Science &Technology, Wuhan, China	MS: Microelectronic Engineering GPA: 88.7/100
09/2012 – 06/2016	Huazhong University of Science &Technology, Wuhan, China	BS: Optoelectronic Information Engineering GPA: 84.7/100

RESEARCH EXPERIENCE

Distance accelerator for Time Series Mining, Advisor: Prof. Yu Hu 11/2016 – 08/2017

- The objective is to develop a high-throughput hardware acceleration technique which can be used for different distance functions in data miming (including dynamic time warping, longest common subsequence, etc.).
- Extacted the basic primitives of these distance functions and split them into two connection structures: matrix structure and row structure.
- Implemented the circuit design for each specific distance function in SPICE with 32nm technology and proposed an early decision algorithm in MATLAB according the indentical feature of analog circuit structure.
- Compared to existing works, the MDA shows an overall speedup of 3.5x-376x on performance and an improvement of 1-3 orders of magnitude on energy efficiency.

Memristor-based 3D Biomedical Image compression, Advisor: Prof. Yu Hu 07/2018 – Present

- The objective is to develop a computation and energy efficient hardware acceleration technique for a 3D Discrete Wavelet Transform(DWT) which is widely used for biomedical image compression.
- Expanded a fast 2D DWT algorithm to 3D version and implemented the 3D level algorithm of DWT in MATLAB.
- Compressed the transformed image by quantization and thresholding, implemented the 3D Inverse Discrete Wavelet Transform(IDWT) in MATLAB.

PUBLICATION

Xiaowei Xu, Feng Lin, Wenyao Xu, Xinwei Yao and Yiyu Shi, **Dewen Zeng**, Yu Hu, "MDA: A Reconfigurable Memristor-based Distance Accelerator for Time Series Mining on Data Centers", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**) (in print). (**corresponding author**)

Xiaowei Xu, **Dewen Zeng**, Wenyao Xu, Yiyu Shi, Yu Hu, "An Efficient Memristor-Based Distance Accelerator for Time Series Data Mining on Data Centers", 54th Design Automation Conference (**DAC**'17), 2017.

INTERNSHIP

Wuhan Easylinkin Technology Co., Ltd. 09/2017 – 03/2018

- The objective of this project is develop a Indoor locating solution for applicaitons like the management of important property or employees' attendance.
- Designed a hardware device depending on a low-power communication protocol called Ibeacon designed by Apple. It can send out information which contains its relative distance to the recevier periodically about 2 times per second.
- Developed a management system by using Spring framework in Java and Mysql database in order to sort and compute real-time data collected from IoT devices. The locating precision of this system can reach around 8 meters with distance error about 0.5 meter.
- Worked on a demonstration system by using Javascript and Css to simulate the real world enviromemt application.

COURSE PROJECT

FPGA-based simple Digital Oscilloscope 03/2015 – 05/2015

- Designed a circuit system integrated a sample-hold circuit, a controllable gain amplifying circuit and a trigger circuit, which can detect sine wave, square wave and triangle wave with the voltage magnitude range from 8mV to 8V.
- Implemented a controlling system based on FPGA DE0-nano platform designed by Altera for generating clock and controlling ADC. Demonstrated the waveform on a TFT screen controlled by Nios II processor on the FPGA.

STM32-based Pulse Oximeter 09/2015 – 11/2015

- Implemented a detection circuit to collect pulse signal from the pulse detection device and a multi-stage amplifying and filtering circuit system for amplifying small pulse signal and eliminate irrelevant signal.
- Investigated a FFT-based low pass filter on a stm32 platform and implemented an algorithm for computing the pulse rate and oxygen saturation. Obtained relative error of around 15% and 20% for pulse rate and oxygen saturation respectively compared to a sophisticated pulse oximeter device.

ADDITIONAL SKILLS

- Windows and Linux operating systems;
- Programming languages: C/C++, Java, Javascript, Python, MATLAB;
- Hardware related: Verilog;
- PCB design using Altium Designer;

AWARDS

- Second prize, National Undergraduate Electronics Design Contest(NUEDC), 2015
- Outstanding undergraduate, Qiming College, Huazhong University of Science and Technology, 2016